Which constitutive model do measured cavitation records prefer?

PyIMR

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Abstract

We compare seventeen constitutive models against laser-induced cavitation records in gelatin at three temperatures, using Bayesian model selection with a noise scale measured from trial-to-trial spread and a prior that prices a model down where it merely reproduces a simpler one it contains. A quadratic Zener solid — strain-stiffening elasticity *and* stress relaxation — wins at every temperature, reaching a residual of $\chi^2/N = 1.23$ to 1.35 against experimental repeatability. Neither ingredient suffices alone: relaxation without stiffening reaches 1.41 to 2.94, and every memoryless model lands above 12. The advantage of the stiffening term collapses as the gel warms, which is the clearest physical signal in the comparison. We report the limitations plainly: the reported residuals are ceilings rather than estimates, because the winning model's best-fitting region is partly unintegrable.

1 Setup

Each dataset is a set of repeated laser-induced cavitation events at one temperature, recorded as $R(t)/R_{\max}$. Figure 1 shows the mean trace and the trial-to-trial spread, which is what we use as the measurement noise: it is measured rather than assumed, so a residual of $\chi^2/N \approx 1$ means a model reproduces the record to within experimental repeatability.

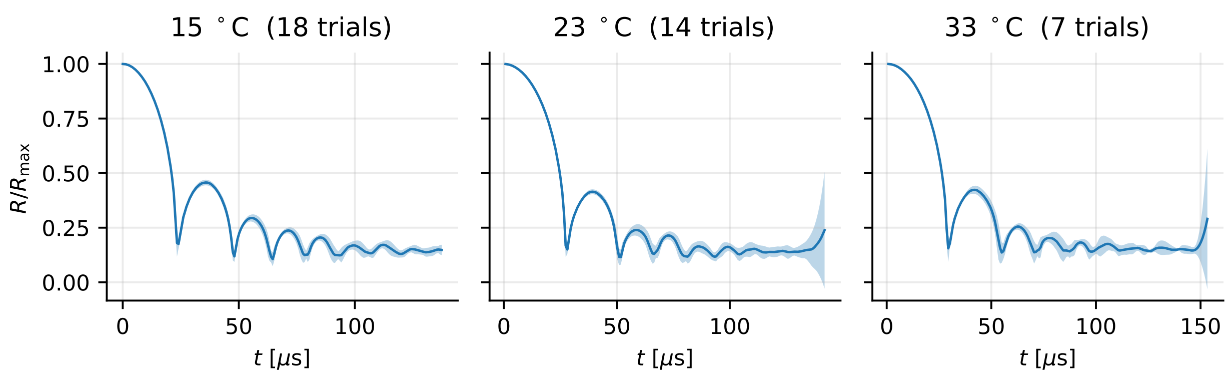

Figure 1: Measured records. Line is the mean over trials, band is ± one standard deviation across trials at each time. The spread is the noise model.

Two samples are excluded before fitting. A trial reading above its own maximum radius is a tracking failure, and the $t = 0$ sample where every trial agrees exactly is a definition rather than a measurement — giving it an error bar would invent an uncertainty and let it constrain the fit.

Each candidate is evaluated on a log-spaced grid over its own parameters, at one resolution for every model so that grid luck cannot decide the winner. The evidence is a grid quadrature over that box, with the noise scale marginalised under a half-Cauchy prior, a redundancy factor that down-weights parameters where a model merely reproduces a simpler model it contains, and a BIC-style complexity penalty. Trajectories are compressible (Keller–Miksis); the incompressible approximation changes the residuals by a factor of two to four and must not be used for these records.

2 Result

Table 1 and Figure 2 give the outcome. The quadratic Zener model `qSLS` — strain-stiffening elasticity with stress relaxation — wins at every temperature, and the ordering beneath it separates cleanly into three tiers: models with both ingredients, models with relaxation only, and memoryless models.

Table 1: Best χ^2/N by model class. Values near 1 indicate a fit within experimental repeatability.

	15 °C	23 °C	33 °C
`qSLS` (stiffening + relaxation)	**1.35**	**1.33**	**1.23**
`SLS` (relaxation, linear elasticity)	2.94	1.85	1.41
viscoelastic fluids	5.3–5.7	4.5–4.7	1.9–2.1
memoryless	12.6–16.7	—	—

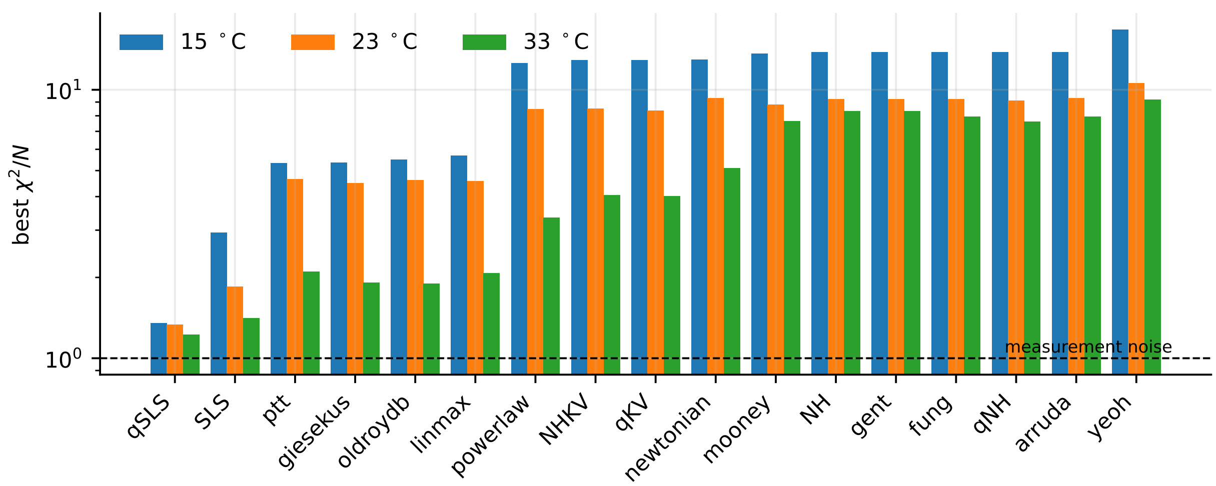

Figure 2: Best χ^2/N for every candidate, all three temperatures, logarithmic scale. The dashed line is the measurement-noise floor.

Figure 3 overlays the winning model on the data it was selected against.

2.1 Reading the posteriors

The model posteriors saturate: the winner takes essentially all the mass and the runners-up are reported as 10^{-29} or smaller. These magnitudes should not be read literally. Evidence differences scale with sample count, and with roughly 3600 samples a modest per-sample difference in fit compounds into an astronomical Bayes factor. The defensible reading is ordinal — the winner is

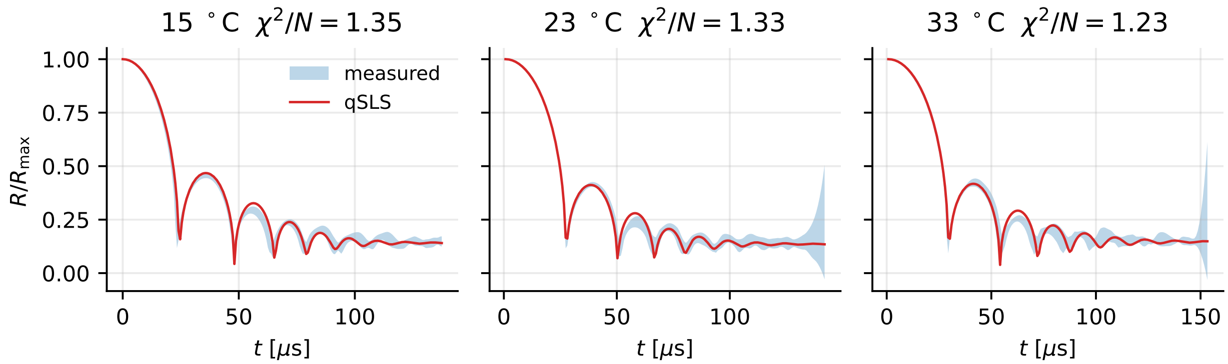

Figure 3: The selected model against the measured band it was fitted to.

decisive and the ordering is meaningful, the magnitudes are not. The residual χ^2/N is the quantity to quote.

3 The temperature trend

The most informative result is not any single number but a trend (Figure 4). The advantage that the stiffening term buys over pure relaxation shrinks monotonically as the gel warms: the gap between `SLS` and `qSLS` falls from 2.94 versus 1.35 at 15 °C to 1.41 versus 1.23 at 33 °C. A warmer, softer gel needs less strain-stiffening to explain its record. Because this is a trend across three independent datasets rather than a single fitted value, it is harder to attribute to a fitting artefact than any individual χ^2.

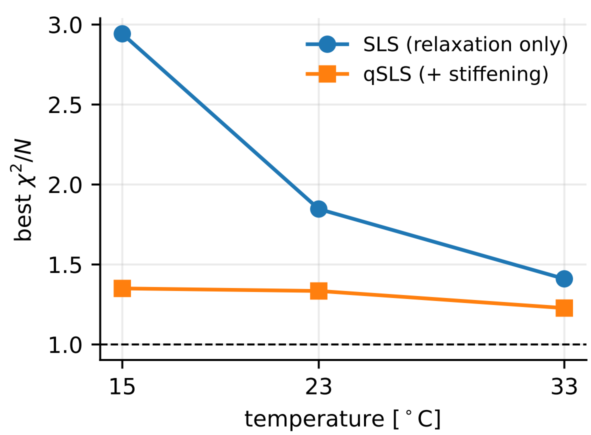

Figure 4: What the stiffening term buys, against temperature. The advantage of `qSLS` over `SLS` narrows as the gel warms.

4 Limitations

The dropped points are outside the model, not outside the solver. Roughly one point in six of the winner's parameter box cannot be integrated. That is not a numerical shortcoming. In that region the model expands to $R/R_0 = 2132$ after its collapse — identically at relative tolerances of 10^{-3}, 10^{-4} and 10^{-5}, so a converged property of the equations rather than an artefact. The bubble begins at its maximum radius and its energy is finite, so growth of that size is the constitutive model failing, and refusing those points is the correct outcome.

The count of physically admissible points is pinned near 1958 of 2401 sampled whatever the settings: looser tolerances and a 20× step budget convert failures into runaways, never into usable solves. This corrects an earlier reading of the same failures as a stiffness problem. It also means the dropped points cannot be recovered by numerical effort, and that no claim about the winner's fit should be made in that region.

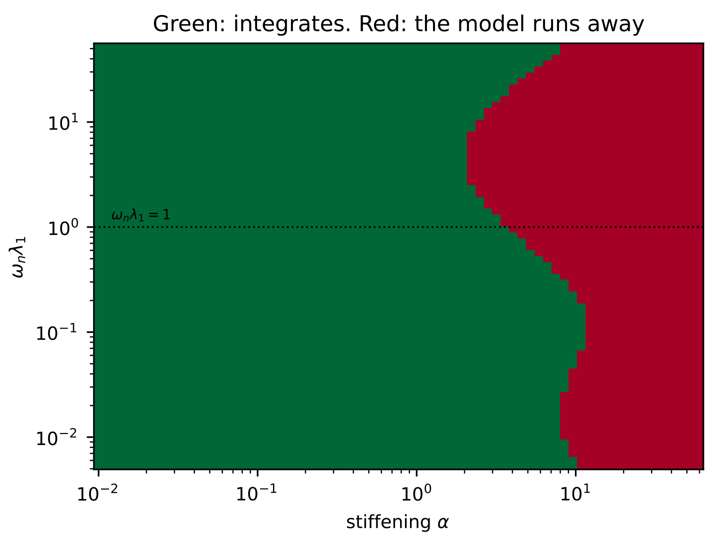

Figure 5: Where the winner can be integrated, at the fitted modulus and viscosity. Failure is governed principally by the stiffening, and the threshold falls to its lowest where the relaxation time is comparable to the bubble's own period — but it is a boundary that moves, not an isolated band.

The ranking survives; the parameter estimates do not. Independent sequential Monte Carlo runs give the winner 2173.55 ± 0.19 in log evidence against 2071.90 ± 0.38 for the relaxation-only model — a gap of 101.6 against run-to-run noise near 0.4, so the comparison is resolved by a wide margin. The tensor grid cannot say this on its own: it returns a single number with no uncertainty.

The same grid cannot estimate parameters. Refining it from 12 to 20 points per axis moves the log evidence by 17.3 and moves the best-fit point from $g = 658$, $\alpha = 1.52$ to $g = 144$, $\alpha = 8.86$, because the best cell is simply whichever node lands nearest the ridge. Refinement cannot fix this: the posterior standard deviations are 0.0018, 0.0025, 0.0055 and 0.078 in unit coordinates against a grid spacing of 0.053, so the posterior occupies about 10^{-7} of the prior volume and a uniform grid would need of order 10^{10} points. Discretisation error of 17.3 sits well inside the 101.6 gap, so the ranking is safe; the reported best-fit parameters are not. Figure 6 shows which models lose points.

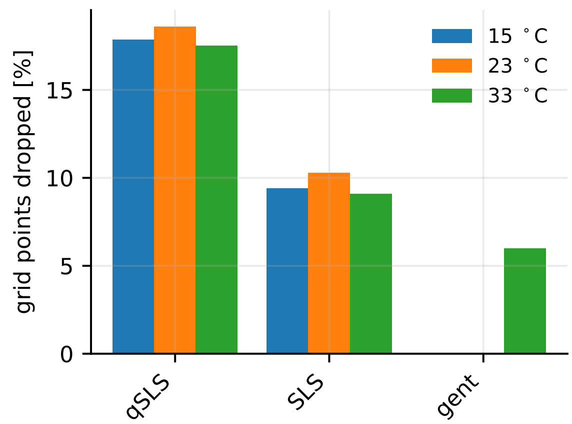

Figure 6: Fraction of each model's parameter grid that could not be integrated. Models not shown lost no points.

5 What the records determine, and what they do not

The winner fits, but not every parameter it contains is measured by these data.

Why the trade exists

Write $\hat{g} = G/p_\infty = 1/\mathrm{Ca}$ for the nondimensional modulus and $\lambda = R_{\mathrm{eq}}/R$ for the stretch the stress law sees. The equilibrium stress the Maxwell arm relaxes toward is

$$Z_e = R^3\left[\tfrac{3\alpha-1}{2}\,\hat{g}\,P(\lambda) + 2\alpha\,\hat{g}\,Q(\lambda)\right], \qquad \begin{aligned} P(\lambda) &= 5 - \lambda^4 - 4\lambda, \\ Q(\lambda) &= \tfrac{27}{40} + \tfrac{1}{8}\lambda^8 + \tfrac{1}{5}\lambda^5 + \lambda^2 - \tfrac{2}{\lambda}. \end{aligned} \tag{1}$$

Collecting powers of α separates it into two terms with different parameter dependence,

$$Z_e = \hat{g}\,\alpha\,A(\lambda) + \hat{g}\,B(\lambda), \qquad A = R^3\left[\tfrac{3}{2}P + 2Q\right], \qquad B = -\tfrac{1}{2}R^3 P. \tag{2}$$

This is exact; it reproduces the implemented expression to 2.5×10^{-14} over random states. Only the first term carries α, so $\hat{g}\alpha$ and $\hat{g}$ are separately determined only while *both* terms contribute.

They do not, at depth. As λ grows, $Q \to \frac{1}{8}\lambda^8$ dominates every other term in either bracket, so $A \to \frac{1}{4}R^3\lambda^8$ while $B \sim \frac{1}{2}R^3\lambda^4$, and

$$Z_e \;\longrightarrow\; \tfrac{1}{4}\,\hat{g}\,\alpha\,R^3\lambda^8, \tag{3}$$

a function of the *product* $\hat{g}\alpha$ alone. The likelihood cannot then distinguish a soft material that stiffens sharply from a stiff one that barely stiffens. Holding Z_e fixed in (2) gives the trade in closed form,

$$\frac{\partial \log \hat{g}}{\partial \log \alpha} = -\frac{\alpha A}{\alpha A + B} \;\longrightarrow\; -1 \quad (\lambda \to \infty), \tag{4}$$

and the share of the signal that separates the two parameters is

$$\frac{|\hat{g}B|}{|\hat{g}\alpha A|} = \frac{|B|}{\alpha|A|} \simeq \frac{2}{\alpha\lambda^4}. \tag{5}$$

Equations (4) and (5) are what the measurements see. These records collapse to $\lambda = 2.215$, where (4) gives -0.981 against -0.99 from the Fisher curvature — a prediction from the constitutive law, not a fit. The profile likelihood returns -0.80 because it averages over a range of λ, and the exponent is -0.861 at $\lambda = 1.5$. Separability falls as λ^{-4}: it is 2% of the signal at $\lambda = 2.2$ and 16% at $\lambda = 1.5$.

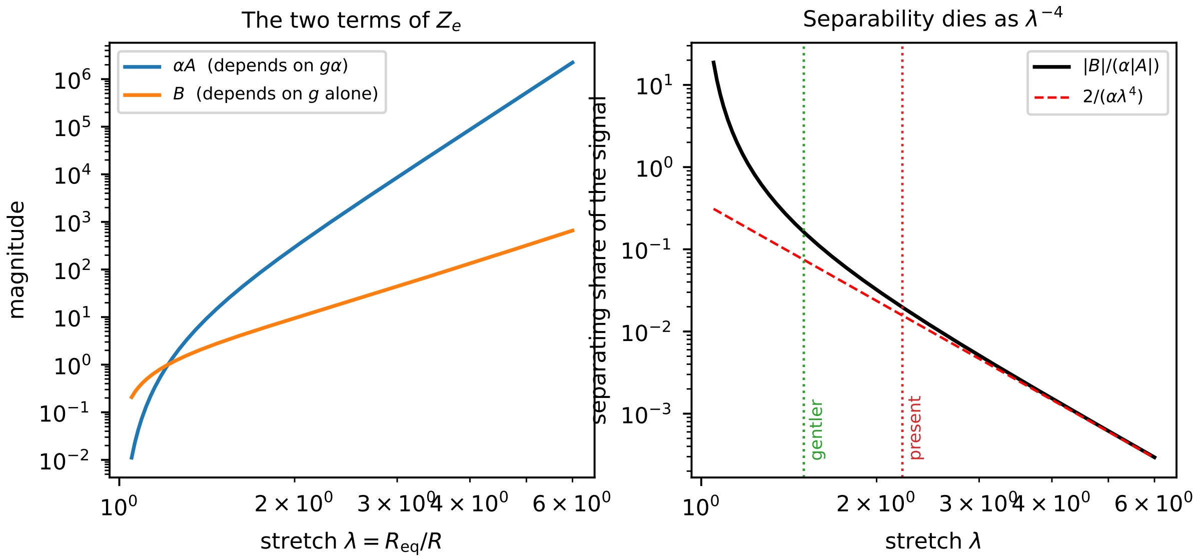

Figure 7: The two terms of (2) against collapse depth. Left: αA, which depends on the product $g\alpha$, outruns B, which carries g alone. Right: their ratio, the share of the signal that can separate the two parameters, following $2/(\alpha\lambda^4)$ once the collapse is deep. It is 2% at the depth these records reach and 16% at $\lambda = 1.5$.

What the Fisher information measures

For Gaussian observations with known deviations the log-likelihood is $-\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2$ up to a constant, so with the whitened sensitivity $J_{ij} = \sigma_i^{-1}\partial m_i/\partial\theta_j$ the Fisher information is exactly $F = J^{\mathsf{T}}J$, and $\Delta\chi^2 \simeq \Delta\theta^{\mathsf{T}} F \Delta\theta$. Taking θ to be the logarithms of the parameters makes the eigenvectors read as power-law combinations: along an eigenvector v the ratio v_g/v_α is the exponent in $g \propto \alpha^{v_g/v_\alpha}$. This is also the observability Gramian of the system augmented with $\dot\theta = 0$, which is why identifiability and observability are the same statement here.

One caution, since it caught us. When the sensitivities are taken with respect to a unit cube $u_j = (\log\theta_j - a_j)/w_j$ rather than $\log\theta_j$ directly, the eigenvector components carry the box widths: $\Delta\log\theta_j = w_j\,\tilde v_j$, so the exponent is $(w_g\tilde v_g)/(w_\alpha\tilde v_\alpha)$, not $\tilde v_g/\tilde v_\alpha$. With w of three and four decades the two differ by 25%.

Designing without assuming the model

Everything above optimises the Fisher information of one model at its fitted parameters, which is circular if that model is wrong. Two checks that do not privilege it.

The record is information-rich; the parameterisation is not. Drawing 400 trajectories from the posterior and taking the singular values of the centred ensemble, eleven modes rise above

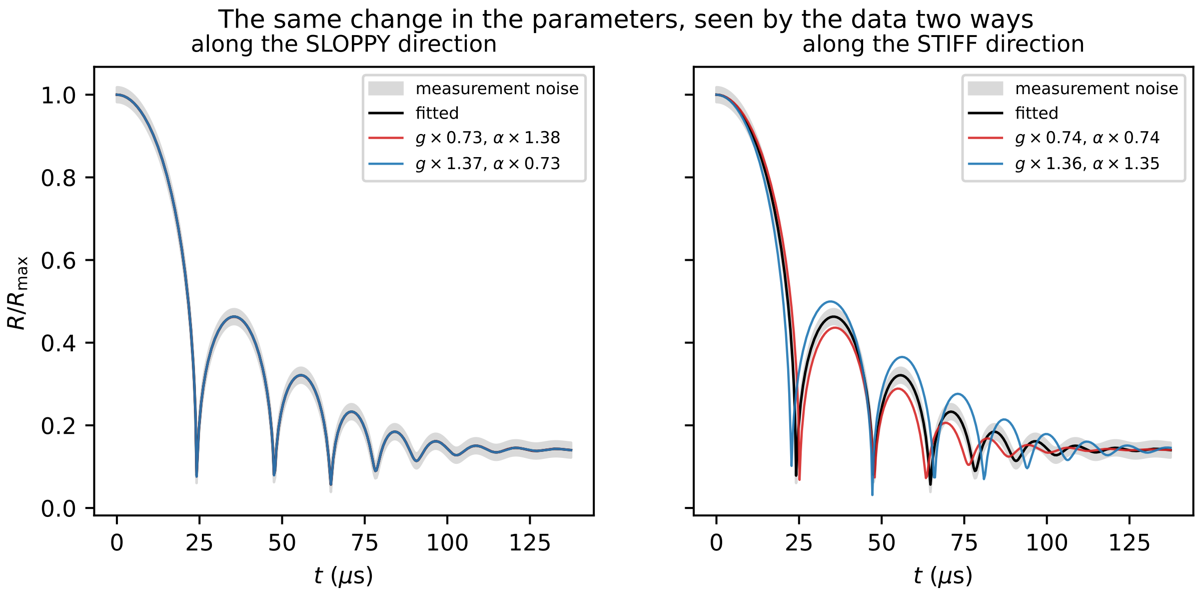

Figure 8: What an eigenvector means, shown rather than defined. The parameters are moved the same distance in log-space along each eigenvector. Along the sloppy direction a 27% change in the modulus, compensated by 38% in the stiffening, leaves the trajectory inside the measurement noise; along the stiff direction a change of the same size is obvious. The data cannot see the first and cannot miss the second.

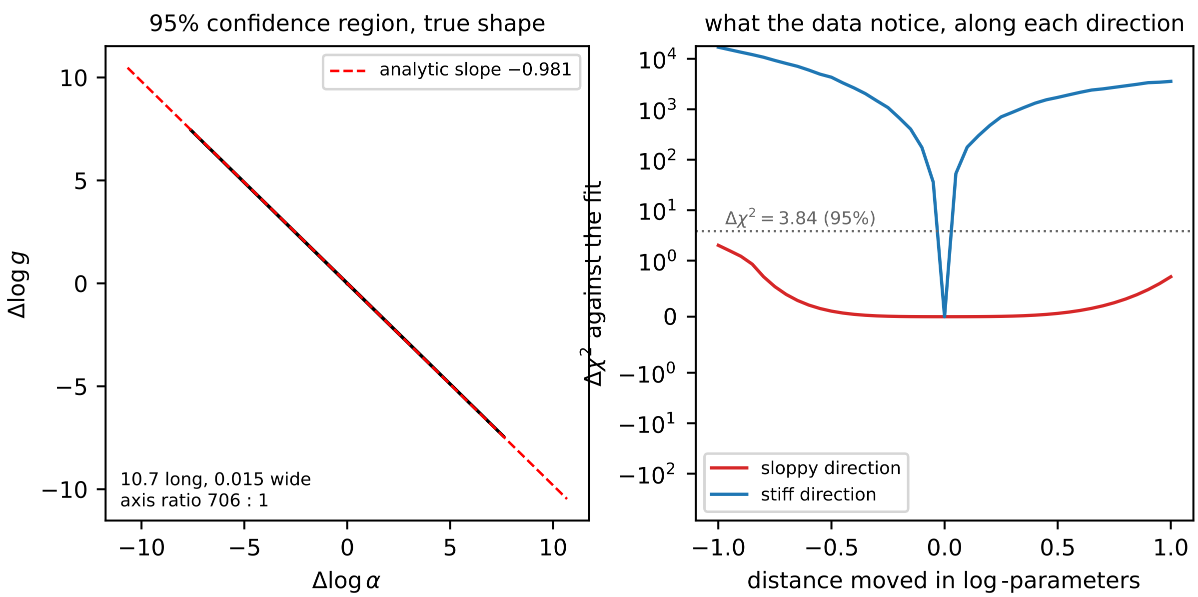

Figure 9: Left: the 95% region from the inverse Fisher information, 10.7 long and 0.015 wide in log-parameters — an axis ratio of 706, which is why it draws as a line, and the analytic slope of -0.981 lies along it. Right: the misfit measured by re-solving at perturbed parameters, not the quadratic approximation. Moving along the stiff direction costs 10^4 in $\Delta\chi^2$; moving the same distance along the sloppy one stays below the 95% threshold, so the parameters may shift by a factor of $e \approx 2.7$ without the data objecting.

the trial noise floor of $0.018\,R_{\max}$. That is not a count of identifiable combinations — four varied parameters span at most a four dimensional manifold, and the extra modes are its curvature — but it does say the measurement can distinguish far more shapes than the fit extracts. The limit is how qSLS parameterises the physics, not the information in the record.

Discrimination does not privilege a model. T-optimality (Atkinson and Fedorov, *Biometrika* **62**, 1975) asks a different question: which design makes the candidates disagree most? Root-mean-square separation between predictions at each model's own fit, in units of the measurement noise:

design	qSLS–SLS	qSLS–qKV	qSLS–NHKV
$277\,\mu$m, stretch 7.09 (present)	2.69	4.92	5.91
$277\,\mu$m, stretch 5.00	5.06	6.78	7.65
$277\,\mu$m, stretch 3.00	5.46	7.51	9.96
$895\,\mu$m, stretch 4.93	**8.08**	8.13	8.55
$100\,\mu$m, stretch 20.0	6.62	8.93	9.19

The present design is the weakest in the set at separating the candidates: qSLS and SLS differ by 2.7 noise units where a gentler collapse gives 5.1 and a larger, gentler one gives 8.1.

This matters because the two objectives share no assumptions. One maximises precision *within* qSLS and presumes it correct; the other measures divergence *between* models and presumes none of them. They point the same way. That agreement is stronger evidence for the recommendation than either alone, and it is the answer to the obvious objection that a model-specific design criterion cannot justify itself.

Where they differ is which gain to prefer. Conditioning within qSLS favours the present bubble size at a gentler collapse; discrimination favours a larger bubble as well. Since the residual analysis says the model is missing structure, discrimination is the more defensible priority, and that argues for the larger bubble.

Why the design criteria disagree

The classical criteria are all functionals of the same F: $\det F$ (D), $\operatorname{tr} F^{-1}$ (A), $\lambda_{\min}$ (E), and $c^{\mathsf{T}} F^{-1} c$ for one combination (c). A design change that merely amplifies the signal — a larger bubble, a longer record, less noise — multiplies F by a scalar s. Then $\lambda_{\min} \to s\lambda_{\min}$ and $\det F \to s^{p} \det F$, so D and E both improve, while

$$\kappa = \lambda_{\max}/\lambda_{\min} \tag{6}$$

is unchanged. Degeneracy is a property of the *directions*, and only a design that alters them can fix it. That is why the E-optimal search preferred a larger bubble ($895\,\mu$m, $\lambda_{\min} = 12.8$) while the conditioning and the eigenvector composition preferred a gentler collapse at the present size ($277\,\mu$m at stretch 5, $\kappa = 4.2 \times 10^{3}$, and the modulus-stiffening share of the worst direction falling from 87% to 66%). Both are correct answers to different questions, and (5) says which question matters here.

Modulus and stiffening trade against each other. Two independent analyses agree. The Fisher information at the fitted optimum has an eigenvalue spectrum spanning six decades, and the eigenvector of its smallest eigenvalue is almost purely a shear-modulus-against-stiffening direction, with weights -0.796 and $+0.604$ and contributions of -0.003 and -0.032 from viscosity and relaxation time. Separately, profiling the likelihood in α — fixing it and re-optimising everything else — gives

$g \propto \alpha^{-0.80}$ across two decades. Sampling the posterior gives a third, independent measurement: its widest direction is $(-0.682, +0.002, -0.082, +0.727)$ in the logarithms of $(g, \mu, \lambda_1, \alpha)$, and $\log g$ is 98.7% anticorrelated with $\log \alpha$.

The three exponents are -0.99 from local curvature, -0.94 from the posterior covariance, and -0.80 from the profile. They agree that the trade is near-reciprocal and disagree in the third digit, which is expected rather than troubling: the valley is curved, so a curvature estimate at one point and a chord measured across two decades are not measuring the same slope. The correlation is the more robust statement.

So viscosity and relaxation time are cleanly determined, while modulus and stiffening are constrained only in combination. The identifiable quantity is closer to $g\,\alpha^{0.8}$ than to either alone, and a fitted α should be reported with its correlation to g rather than as an independent material property.

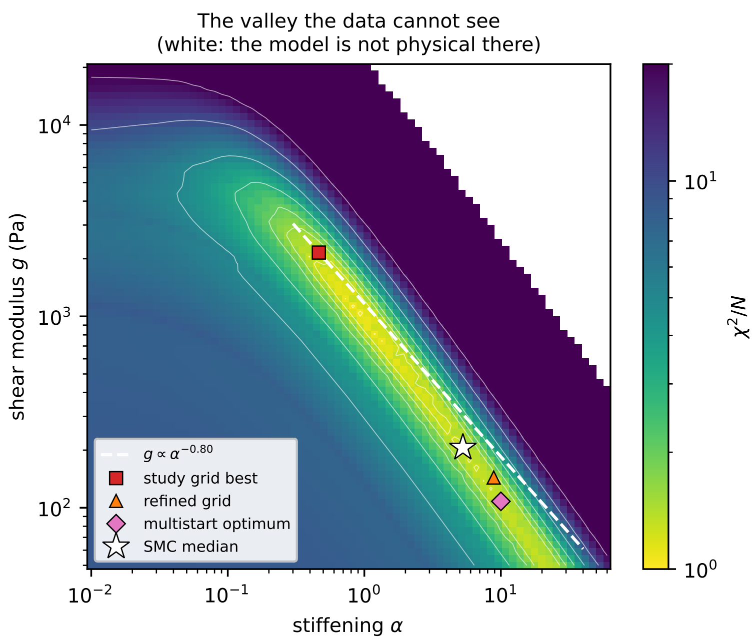

Figure 10: χ^2/N over the modulus–stiffening plane, at the fitted viscosity and relaxation time. The dashed line is $g \propto \alpha^{-0.80}$, predicted from the Fisher curvature at a single point; it lies along the floor of the valley. Four "best fits" from four methods are strung out along that floor rather than disagreeing with one another. The white region at upper right is where the model is not physical.

Figure 10 shows why the four estimates in this study disagree about g by an order of magnitude while agreeing about the fit: they are different points on the floor of a single flat-bottomed trench.

The stiffening is nonetheless identified, once the prior allows it. Sampling with the ceiling on α placed at 1, 10 and 100:

ceiling	log evidence	median α	97.5%
1	2162.9	0.85	0.97
10	2173.9	6.37	9.04
100	2172.4	5.27	8.73

Widening the prior tenfold moves the median only from 6.37 to 5.27 and the evidence by 1.5 out of 2173, so α is identified near 5 with 97.5% of posterior mass below about 9. A prior-dominated parameter would have tracked its bound. The ceiling of 1 shows what genuine truncation looks like: the median pins against it and 11 log units of evidence are lost.

This corrects a natural reading of the profile likelihood alone, which appears flat to the boundary only because the boundary sits where the posterior's upper tail does.

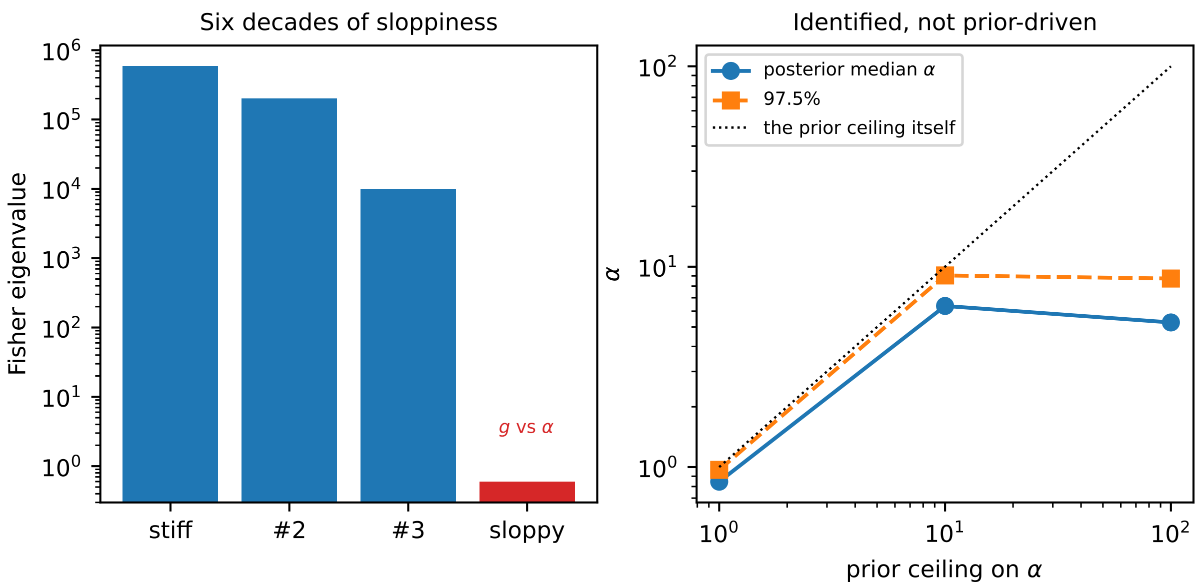

Figure 11: Left: the Fisher spectrum at the optimum, six decades between the stiffest and sloppiest directions, with the sloppiest being modulus against stiffening. Right: the posterior for α against the ceiling imposed on its prior. It tracks the ceiling while truncated and departs from it once the ceiling is loose, which is what distinguishes an identified parameter from a prior-driven one.

Consequences for the numbers reported here. Three methods that resolve the posterior — sampling, a refined grid, and continuous optimisation — agree that g lies near 10^2 Pa. The coarse grid used for the ranking reports 2154 Pa. The ranking does not depend on this, but any material property quoted from the grid does. We also note that a modulus of that size is implausibly soft for gelatin, which is the degeneracy above making itself felt: the fit buys a lower residual by trading modulus against stiffening, along a direction the data barely constrain.

One candidate was never genuinely exercised. The Gent solid appears in the comparison but cannot be tested on these records. Its energy diverges as $I_1 - 3 \to J_m$, and a Gent solid soft enough to be distinguishable from Neo-Hookean predicts a collapse it cannot sustain, so it locks up. Where it does integrate, it differs from Neo-Hookean by 1.7×10^{-4} of $R_{\max}$ against measurement noise near 2×10^{-2} — two orders of magnitude below what the data can resolve. Any statement about strain-stiffening laws here excludes it.

The noise model sees only repeatability. Using trial-to-trial spread as the noise means $\chi^2/N \approx 1$ certifies agreement to within experimental repeatability. A systematic error common to every trial would not appear in that spread, and would show up instead as exactly the kind of residual excess we report — 20 to 35% above the noise floor. This study cannot separate that from genuine model inadequacy.

6 Reproducing

```
.venv/bin/python examples/measured_selection.py <dataset> [thermal Nt] [workers]
.venv/bin/python docs/writeup/collect.py 16     # records results.json
.venv/bin/python docs/writeup/figures.py        # draws every figure from it
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Every number and figure in this document comes from `results.json`; none is transcribed by hand.